

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

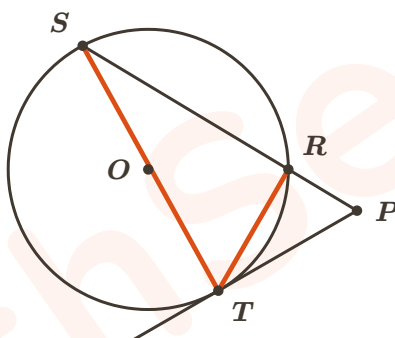
## Chapter 18: Tangents and Intersecting Chords

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

### 1 Tangent and secant from an external point

1.  $PT$  is a tangent to a circle at  $T$ , and a line through  $P$  cuts the circle at  $R$  and  $S$ , with  $R$  nearer to  $P$ . (a) Prove that  $\triangle PTR \sim \triangle PST$ . (b) Prove that  $PT^2 = PR \times PS$ . (c) If  $PR = 4$  cm and  $PS = 16$  cm, find  $PT$ . **ICSE 2025, 3 marks**



SOLUTION

(a) In  $\triangle PTR$  and  $\triangle PST$ :  $\angle P$  is common, and  $\angle PTR = \angle PST$  (angle between tangent and chord equals the angle in the alternate segment). So  $\triangle PTR \sim \triangle PST$  (AA).

(b) Corresponding sides are proportional:  $\frac{PT}{PS} = \frac{PR}{PT}$ , so  $PT^2 = PR \times PS$ .

(c)  $PT^2 = 4 \times 16 = 64$

**Answer:**  $PT = 8$  cm

2. A circle has centre  $O$  and radius 9 cm. The diameter  $AB$  is produced to a point  $P$ , and  $PQ$  is a tangent from  $P$ . If  $PA = 24$  cm, find  $PQ$ . **ICSE 2024, 3 marks**

SOLUTION

$AB = 18$  cm, so  $PB = PA - AB = 24 - 18 = 6$  cm.

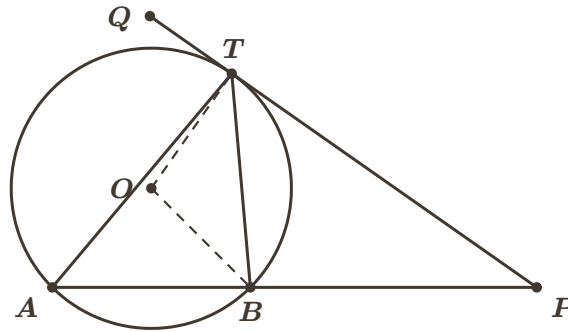
$PQ^2 = PA \times PB = 24 \times 6 = 144$

**Answer:**  $PQ = 12$  cm

(Check:  $OP = 15$  cm and  $OQ = 9$  cm, so  $PQ = \sqrt{15^2 - 9^2} = 12$  cm.)

3.  $PQ$  is a tangent to a circle with centre  $O$  at  $T$ . The chord  $AB$  produced meets the tangent at  $P$ .  $AB = 9$  cm,  $BP = 16$  cm,  $\angle PTB = 50^\circ$  and  $\angle OBA = 45^\circ$ . Find (a)  $PT$  (b)  $\angle BAT$  (c)  $\angle BOT$  (d)  $\angle ABT$ .

ICSE 2023, 4 marks



SOLUTION

(a)  $PA = 9 + 16 = 25$  cm.  $PT^2 = PB \times PA = 16 \times 25 = 400$ , so  $PT = 20$  cm.

(b) By the alternate segment theorem,  $\angle BAT = \angle PTB = 50^\circ$ .

(c) Angle at the centre is twice the angle at the circumference:  $\angle BOT = 2 \times 50^\circ = 100^\circ$ .

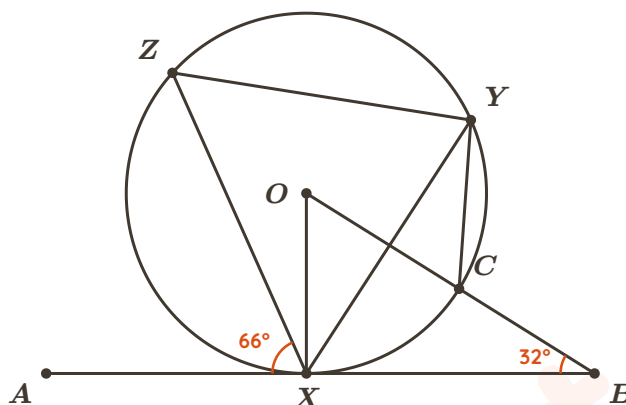
(d)  $OB = OT$ , so  $\angle OBT = \frac{180^\circ - 100^\circ}{2} = 40^\circ$ .  $\angle ABT = \angle OBA + \angle OBT = 45^\circ + 40^\circ = 85^\circ$ .

**Answer: (a) 20 cm (b)  $50^\circ$  (c)  $100^\circ$  (d)  $85^\circ$**

## 2 The alternate segment theorem

4.  $X, Y, Z$  and  $C$  lie on a circle with centre  $O$ .  $AB$  is the tangent at  $X$ ,  $OB$  cuts the circle at  $C$ , and  $ZY = XY$ . Given  $\angle OBX = 32^\circ$  and  $\angle AXZ = 66^\circ$ , find (a)  $\angle BOX$  (b)  $\angle CYX$  (c)  $\angle ZYX$  (d)  $\angle OXY$ .

ICSE 2025, 4 marks



SOLUTION

(a)  $OX \perp AB$  (radius and tangent). In  $\triangle OXB$ :  $\angle BOX = 90^\circ - 32^\circ = 58^\circ$ .

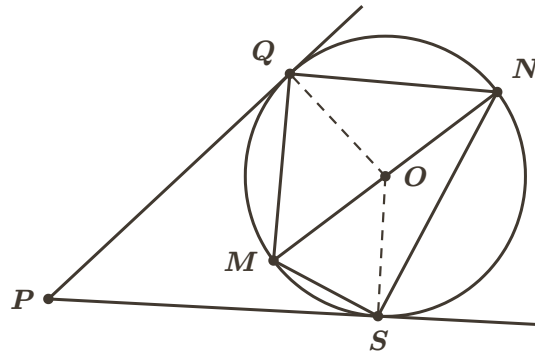
(b) The arc  $CX$  subtends  $\angle COX = 58^\circ$  at the centre and  $\angle CYX$  on the circle:  $\angle CYX = \frac{58^\circ}{2} = 29^\circ$ .

(c) By the alternate segment theorem,  $\angle ZYX = \angle AXZ = 66^\circ$ .

(d)  $ZY = XY$ , so  $\angle YXZ = \frac{180^\circ - 66^\circ}{2} = 57^\circ$ . Also  $\angle OXZ = 90^\circ - 66^\circ = 24^\circ$ . So  $\angle OXY = 57^\circ - 24^\circ = 33^\circ$ .

**Answer: (a)  $58^\circ$  (b)  $29^\circ$  (c)  $66^\circ$  (d)  $33^\circ$**

5.  $PR$  and  $PT$  are tangents from  $P$  to a circle with centre  $O$ , touching it at  $Q$  and  $S$ .  $MN$  is a diameter, with  $M$  on the arc  $QS$  nearer to  $P$ . Given  $\angle PQM = 42^\circ$  and  $\angle PSM = 25^\circ$ , find (a)  $\angle OQM$  (b)  $\angle QNS$  (c)  $\angle QOS$  (d)  $\angle QMS$ . **ICSE 2024, 4 marks**



SOLUTION

(a)  $OQ \perp PR$  (radius and tangent), so  $\angle OQM = 90^\circ - 42^\circ = 48^\circ$ .

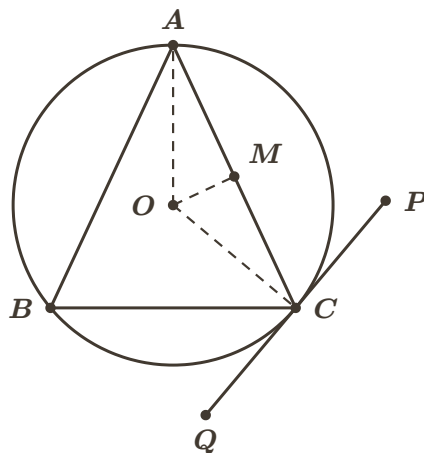
(b) By the alternate segment theorem,  $\angle QNM = \angle PQM = 42^\circ$  and  $\angle SNM = \angle PSM = 25^\circ$ . So  $\angle QNS = 42^\circ + 25^\circ = 67^\circ$ .

(c) Angle at the centre is twice the angle at the circumference:  $\angle QOS = 2 \times 67^\circ = 134^\circ$ .

(d)  $QNSM$  is a cyclic quadrilateral:  $\angle QMS = 180^\circ - 67^\circ = 113^\circ$ .

**Answer: (a)  $48^\circ$  (b)  $67^\circ$  (c)  $134^\circ$  (d)  $113^\circ$**

6. An isosceles triangle  $ABC$  with  $AB = AC$  is inscribed in a circle with centre  $O$ .  $PQ$  is the tangent at  $C$ , with  $Q$  on the side of  $BC$  away from  $A$ .  $OM$  is perpendicular to the chord  $AC$  and  $\angle COM = 65^\circ$ . Find (a)  $\angle ABC$  (b)  $\angle BAC$  (c)  $\angle BCQ$ . **ICSE 2024, 3 marks**



SOLUTION

- (a) The perpendicular from the centre bisects the chord and the angle at the centre:  $\angle AOC = 2 \times 65^\circ = 130^\circ$ . So  $\angle ABC = \frac{130^\circ}{2} = 65^\circ$ .  
 (b)  $AB = AC$ , so  $\angle ACB = \angle ABC = 65^\circ$  and  $\angle BAC = 180^\circ - 130^\circ = 50^\circ$ .  
 (c) By the alternate segment theorem,  $\angle BCQ = \angle BAC = 50^\circ$ .

**Answer: (a)  $65^\circ$  (b)  $50^\circ$  (c)  $50^\circ$**

7.  $RT$  is a tangent to a circle at  $S$ .  $P$  and  $Q$  are points on the circle with  $\angle PST = 30^\circ$  and  $\angle SPQ = 60^\circ$ . Find  $\angle PSQ$ . (a)  $40^\circ$  (b)  $30^\circ$  (c)  $60^\circ$  (d)  $90^\circ$  **ICSE 2023, 1 mark**

SOLUTION

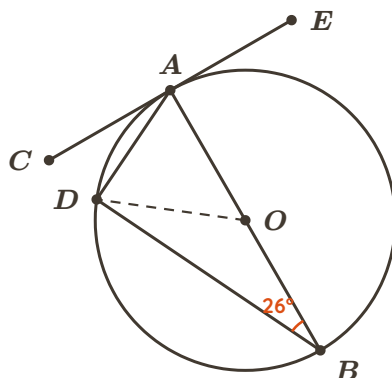
By the alternate segment theorem,  $\angle PQS = \angle PST = 30^\circ$ .

In  $\triangle PQS$ :  $\angle PSQ = 180^\circ - 60^\circ - 30^\circ = 90^\circ$

**Answer: (d)**

8.  $AB$  is a diameter of a circle with centre  $O$ , and  $CE$  is the tangent at  $A$ .  $D$  is a point on the circle on the same side of  $AB$  as  $C$ , with  $\angle ABD = 26^\circ$ . Find (a)  $\angle BDA$  (b)  $\angle BAD$  (c)  $\angle CAD$  (d)  $\angle ODB$ .

ICSE 2023, 4 marks



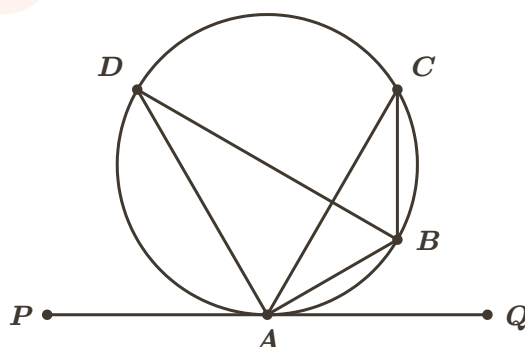
SOLUTION

- (a)  $\angle BDA = 90^\circ$  (angle in a semicircle).  
 (b)  $\angle BAD = 180^\circ - 90^\circ - 26^\circ = 64^\circ$   
 (c) By the alternate segment theorem,  $\angle CAD = \angle ABD = 26^\circ$ .  
 (d)  $OD = OB$  (radii), so  $\angle ODB = \angle OBD = 26^\circ$ .

**Answer: (a)  $90^\circ$  (b)  $64^\circ$  (c)  $26^\circ$  (d)  $26^\circ$**

9.  $PQ$  is a tangent to a circle at  $A$ .  $B, C$  and  $D$  are points on the circle such that  $AB$  bisects  $\angle CAQ$  and  $AD$  bisects  $\angle PAC$ . If  $\angle BAQ = 30^\circ$ , prove that (a)  $BD$  is a diameter (b)  $\triangle ABC$  is isosceles.

ICSE 2017, 3 marks



SOLUTION

- (a)  $\angle CAQ = 2 \times 30^\circ = 60^\circ$ , so  $\angle BAC = 30^\circ$  and  $\angle PAC = 180^\circ - 60^\circ = 120^\circ$ .  
 $AD$  bisects  $\angle PAC$ , so  $\angle CAD = 60^\circ$ . Then  $\angle BAD = \angle BAC + \angle CAD = 30^\circ + 60^\circ = 90^\circ$ .  
 A chord that subtends a right angle at the circumference is a diameter, so  $BD$  is a diameter.  
 (b) By the alternate segment theorem,  $\angle ACB = \angle BAQ = 30^\circ$ . Also  $\angle BAC = 30^\circ$ .  
 Two angles of  $\triangle ABC$  are equal, so  $AB = BC$ . **Both results are proved.**

### 3 Tangent properties

10.  $PS$  and  $PT$  are tangents to a circle from  $P$ , with  $\angle SPT = 80^\circ$ .  $Q$  is a point on the circle with  $SQ \parallel PT$ . Find  $\angle QST$ . (a)  $140^\circ$  (b)  $90^\circ$  (c)  $80^\circ$  (d)  $50^\circ$  **ICSE 2024, 1 mark**

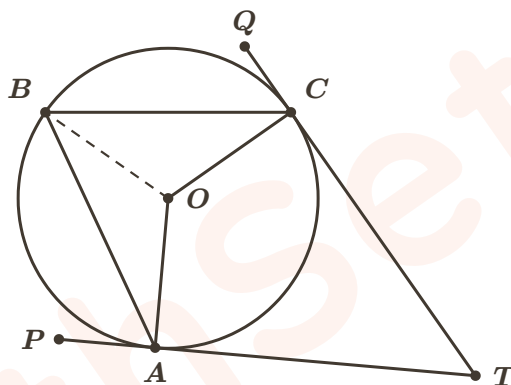
SOLUTION

$$PS = PT, \text{ so } \angle PTS = \angle PST = \frac{180^\circ - 80^\circ}{2} = 50^\circ.$$

$SQ \parallel PT$  and  $ST$  is a transversal, so  $\angle QST = \angle PTS = 50^\circ$  (alternate angles).

**Answer: (d)**

11.  $TP$  and  $TQ$  are tangents to a circle with centre  $O$ , touching it at  $A$  and  $C$ .  $B$  is a point on the circle with  $\angle BCQ = 55^\circ$  and  $\angle BAP = 60^\circ$ . Find (a)  $\angle OBA$  and  $\angle OBC$  (b)  $\angle AOC$  (c)  $\angle ATC$ . **ICSE 2020, 4 marks**



SOLUTION

(a)  $OA \perp TP$ , so  $\angle OAB = 90^\circ - 60^\circ = 30^\circ$ .  $OA = OB$ , so  $\angle OBA = 30^\circ$ .

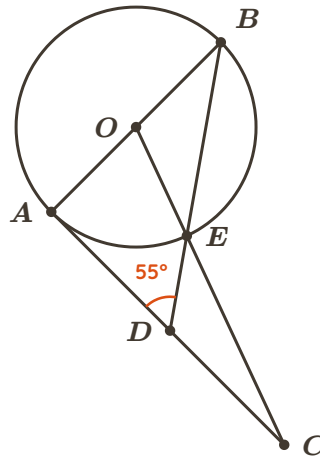
$OC \perp TQ$ , so  $\angle OCB = 90^\circ - 55^\circ = 35^\circ$ .  $OB = OC$ , so  $\angle OBC = 35^\circ$ .

(b)  $\angle ABC = 30^\circ + 35^\circ = 65^\circ$ , so  $\angle AOC = 2 \times 65^\circ = 130^\circ$ .

(c) In quadrilateral  $OATC$  the angles at  $A$  and  $C$  are  $90^\circ$ , so  $\angle ATC = 180^\circ - 130^\circ = 50^\circ$ .

**Answer: (a)  $30^\circ$  and  $35^\circ$  (b)  $130^\circ$  (c)  $50^\circ$**

12.  $AB$  is a diameter of a circle with centre  $O$ , and  $AC$  is the tangent at  $A$ .  $D$  is a point on  $AC$ , and  $BD$  cuts the circle at  $E$ . The line  $OE$  produced meets the tangent at  $C$ . If  $\angle ADB = 55^\circ$ , find  $x = \angle ACO$  and  $y = \angle AOE$ . **ICSE 2019, 3 marks**



**SOLUTION**

$\angle BAD = 90^\circ$  (the tangent is perpendicular to the radius at the point of contact).

In  $\triangle ABD$ :  $\angle ABD = 180^\circ - 90^\circ - 55^\circ = 35^\circ$ .

$y = \angle AOE = 2 \times \angle ABE = 70^\circ$  (the angle at the centre is twice the angle at the circumference).

In  $\triangle OAC$ :  $x = 180^\circ - 90^\circ - 70^\circ = 20^\circ$ .

**Answer:**  $x = 20^\circ, y = 70^\circ$

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